



**SIMATS Online Programming Lab**  
DS PROGRAMMING

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QUESTIONS 4 / 5 completed

Insertion Sort

**INSERTION SORT**  
Write a program sort the elements 9, 5, 1, 4, 3 using Insertion Sort.

**PROGRAM EDITOR**

C

Run Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int i,j,t,a[100],n;
5     scanf("%d",&n);
6     for(i=0;i<n;i++)
7         scanf("%d",&a[i]);
8
9     for(i=1;i<n;i++)
10    {
11        j=i;
12        while(j>0 && a[j]<a[j-1])
13        {
14            t=a[j];
15            a[j]=a[j-1];
16            a[j-1]=t;
17            j--;
18        }
19    }
20    for(i=0;i<n;i++)
21        printf("%d ",a[i]);
```

**INPUT**  
5  
9 5 1 4 3



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using Insertion Sort.

```
10 {
11     j=i;
12     while(j>0 && a[j]<a[j-1])
13     {
14         t=a[j];
15         a[j]=a[j-1];
16         a[j-1]=t;
17         j--;
18     }
19 }
20 for(i=0;i<n;i++)
21     printf("%d ",a[i]);
22 return 0;
23 }
```

**OUTPUT**  
1 3 4 5 9